

# Lecture 15/Module 2

Transient Analysis of First and Second Order Circuit

22/08/2026

Dr. Sudipta Kumar Sarkar

*Assistant Professor (Research), SENSE*



School of Electronics Engineering (SENSE)  
VIT, Vellore, India

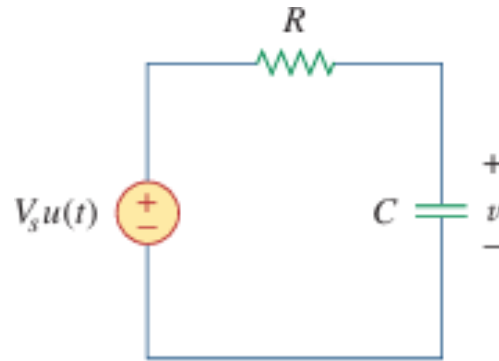
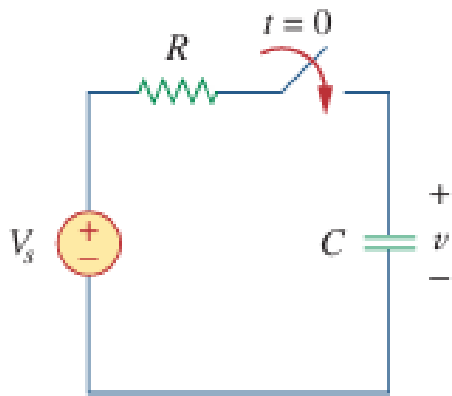
# Today's Topics

- Step Response – RC Circuit
- Exercise

# Step Response of RC circuit

The step response of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.

When the dc source of an RC circuit is suddenly applied, the voltage or current source can be modeled as a step function.



Initial voltage on the capacitor is  $V_0$

$$v(0^-) = v(0^+) = V_0$$

Where  $v(0^-)$  is the voltage across the capacitor just before switching and  $v(0^+)$  is its voltage immediately after switching.

Applying KCL;

$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0$$

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC} u(t)$$

Where  $v$  is the voltage across the capacitor. For  $t > 0$ ,

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$

# Step Response of RC circuit

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$

Rearranging terms gives,

$$\frac{dv}{dt} = -\frac{v - V_s}{RC}$$

$$\frac{dv}{v - V_s} = -\frac{dt}{RC}$$

Integrating both sides and introducing the initial conditions,

$$\ln(v - V_s) \Big|_{V_0}^{v(t)} = -\frac{t}{RC} \Big|_0^t$$

$$\ln(v(t) - V_s) - \ln(V_0 - V_s) = -\frac{t}{RC} + 0$$

$$\ln \frac{v - V_s}{V_0 - V_s} = -\frac{t}{RC}$$

# Step Response of RC circuit

Taking the exponential of both sides,

$$\frac{v - V_s}{V_0 - V_s} = e^{-t/\tau}, \quad \tau = RC$$
$$v - V_s = (V_0 - V_s)e^{-t/\tau}$$

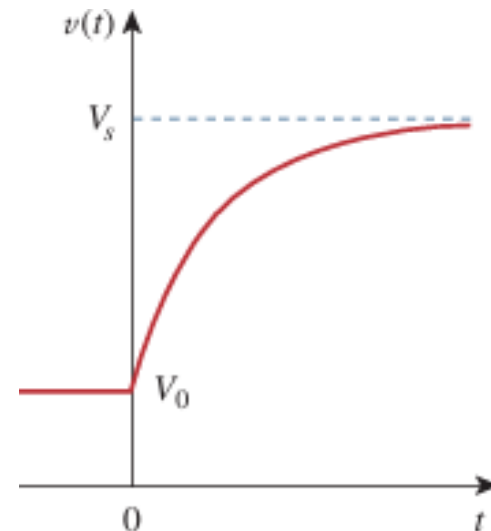
$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$$

Thus,

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}$$

Assuming that  $V_s > V_0$  a plot of  $v(t)$  is shown:

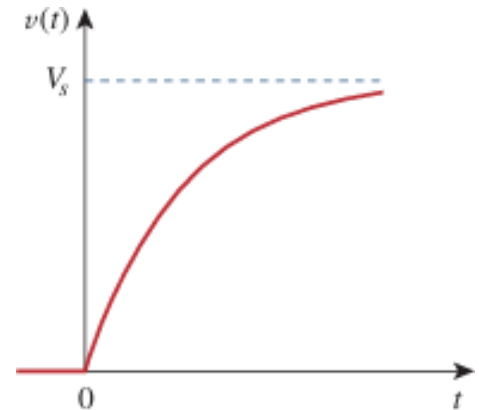
This is known as the **complete response** (or total response) of the RC circuit to a sudden application of a dc voltage source, **assuming the capacitor is initially charged**.



# If $V_o = 0$ , Step Response of RC circuit

$$v(t) = \begin{cases} 0, & t < 0 \\ V_s(1 - e^{-t/\tau}), & t > 0 \end{cases}$$

this can be written alternatively as,  $v(t) = V_s(1 - e^{-t/\tau})u(t)$

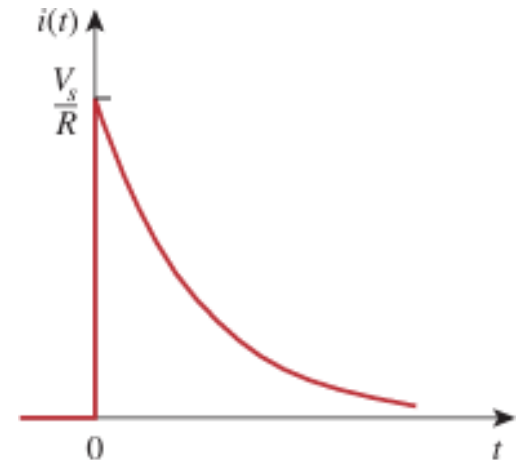


This is the **complete step response** of the RC circuit when the **capacitor is initially uncharged**.

The current through the capacitor is,

$$i(t) = C \frac{dv}{dt} = \frac{C}{\tau} V_s e^{-t/\tau}, \quad \tau = RC, \quad t > 0$$

$$i(t) = \frac{V_s}{R} e^{-t/\tau} u(t)$$



Complete response = natural response + forced response  
stored energy                      independent source

$$v = v_n + v_f$$

(based on the sources of response)

$$v_n = V_o e^{-t/\tau}$$

$$v_f = V_s(1 - e^{-t/\tau})$$

# Step Response of RC circuit

Another way of looking at the complete response is,

$$\text{Complete response} = \underset{\text{temporary part}}{\text{transient response}} + \underset{\text{permanent part}}{\text{steady-state response}}$$

(based on the permanency of response)

$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$$

$$v = v_t + v_{ss}$$

$$v_t = (V_0 - V_s)e^{-t/\tau} \quad v_{ss} = V_s$$

The transient response  $v_t$  is temporary; it is the portion of the complete response that decays to zero as time approaches infinity.

The steady-state response  $V_{ss}$  is the portion of the complete response that remains after the transient response has died out.

In general,

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

where  $v(0)$  is the initial voltage at  $t=0^+$  and  $v(\infty)$  is the final or steady state value.

# Step Response of RC circuit

If the switch changes position at time  $t = t_0$  instead of  $t = 0$  there is a time delay in the response so that the equation becomes,

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

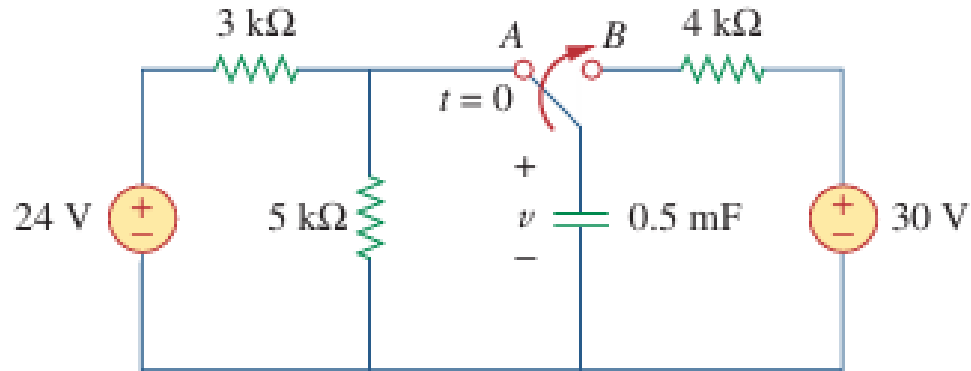
Where  $v(t_0)$  is the initial value at  $t = t_0^+$

Instead of

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

# Exercise

1. The switch in the circuit below has been in position A for a long time. At  $t = 0$  the switch moves to B. Determine  $v(t)$  for  $t > 0$  and calculate its value at  $t = 1$  s and 4 s.



For the time  $t < 0$ , switch is at position A, so  $v_c$  will be the same as drop across  $5\text{ k}\Omega$  resistor.

$$v(0^-) = \frac{5}{5 + 3}(24) = 15\text{ V}$$

$$v(0) = v(0^-) = v(0^+) = 15\text{ V}$$

For the time  $t > 0$ , switch is at position B,  $R_{th} = 4\text{ k}\Omega$

$$\tau = R_{Th}C = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2\text{ s} \quad \text{and} \quad v(\infty) = 30\text{ V}.$$

# Exercise

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

$$= 30 + (15 - 30)e^{-t/2} = (30 - 15e^{-0.5t}) \text{ V}$$

At  $t = 1$ ,

$$v(1) = 30 - 15e^{-0.5} = 20.9 \text{ V}$$

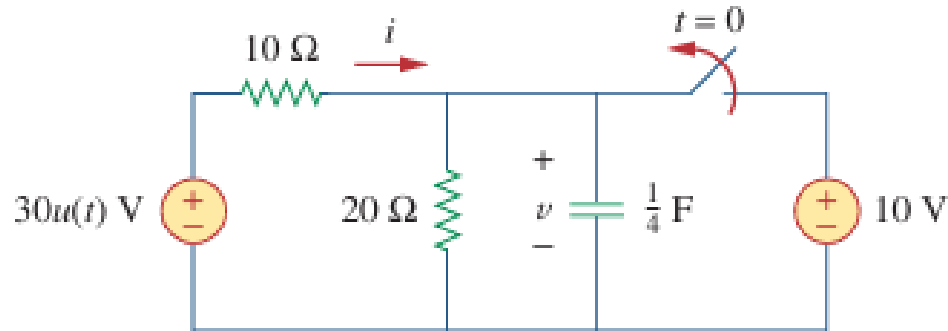
At  $t = 4$ ,

$$v(4) = 30 - 15e^{-2} = 27.97 \text{ V}$$

# Exercise

In the circuit given below, the switch has been closed for a long time and is opened at  $t = 0$ .

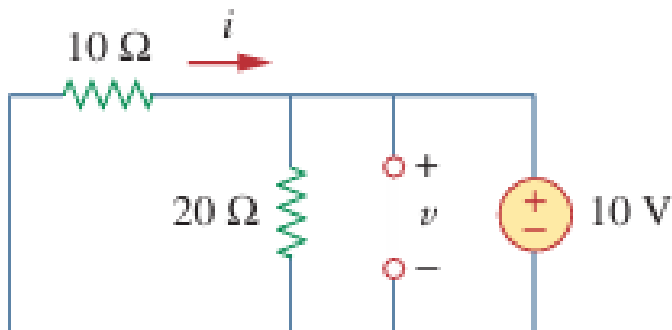
Find  $i$  and  $v$  for all time.



By definition of the unit step function,

$$30u(t) = \begin{cases} 0, & t < 0 \\ 30, & t > 0 \end{cases}$$

For  $t < 0$ , the switch is closed and so that the voltage source is replaced by a short circuit and should be regarded as contributing nothing to  $v$ .

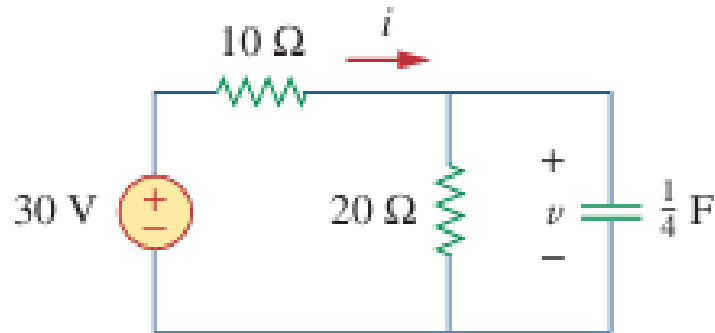


$$v = 10 \text{ V}, \quad i = -\frac{v}{10} = -1 \text{ A}$$

$$v(0) = v(0^-) = 10 \text{ V}$$

# Exercise

For  $t > 0$ , the switch is opened and the 10-V voltage source is disconnected from the circuit.



$$v(\infty) = \frac{20}{20 + 10}(30) = 20 \text{ V}$$

$$R_{\text{Th}} = 10 \parallel 20 = \frac{10 \times 20}{30} = \frac{20}{3} \Omega$$

$$\tau = R_{\text{Th}}C = \frac{20}{3} \cdot \frac{1}{4} = \frac{5}{3} \text{ s}$$

$$\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} \\ &= 20 + (10 - 20)e^{-(3/5)t} = (20 - 10e^{-0.6t}) \text{ V} \end{aligned}$$

$$\begin{aligned} i &= \frac{v}{20} + C \frac{dv}{dt} \\ &= 1 - 0.5e^{-0.6t} + 0.25(-0.6)(-10)e^{-0.6t} = (1 + e^{-0.6t}) \text{ A} \end{aligned}$$

$$v = \begin{cases} 10 \text{ V}, & t < 0 \\ (20 - 10e^{-0.6t}) \text{ V}, & t \geq 0 \end{cases}$$

$$i = \begin{cases} -1 \text{ A}, & t < 0 \\ (1 + e^{-0.6t}) \text{ A}, & t > 0 \end{cases}$$

Thank You